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# INDIAN INSTITUTE OF INFORMATION TECHNOLOGY DHARWAD

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## M.Tech DSAI Courses:

| M.Tech in DSAI Courses |                                                        |   |                        |                 |
|------------------------|--------------------------------------------------------|---|------------------------|-----------------|
| Sl no.                 | Course Names                                           |   | Credit                 | Type            |
| SEM 1                  |                                                        |   |                        |                 |
| 1                      | Applied Mathematics for Data Science and AI            |   | 3                      | Discipline Core |
| 2                      | Machine Learning Paradigms                             |   | 3                      | Discipline Core |
| 3                      | Data Science Lab                                       |   | 2                      | Discipline Core |
| 4                      | Introduction to Speech and Natural Language Processing | 1 | 2<br>(Any two courses) | Elective        |
| 5                      | Introduction to Computer Vision                        | 1 |                        | Elective        |
| 6                      | Introduction to Generative AI                          | 1 |                        | Elective        |
| 7                      | Introduction to Financial Analytics                    | 1 |                        | Elective        |
| 8                      | Introduction to AI in Healthcare                       | 1 |                        | Elective        |
| 9                      | Introduction to Research                               |   | 2                      | Master's Core   |
| 10                     | Project-I                                              |   | 3                      | Project         |
|                        | Total                                                  |   | 15                     |                 |
| SEM 2                  |                                                        |   |                        |                 |
| 7                      | AI Systems Lab                                         |   | 2                      | Discipline Core |
| 8                      | Literature Review and Seminar                          |   | 2                      | Master's Core   |
| 9                      | Electives (1/2/3/4 credits)                            |   | 5                      | Elective        |
| 10                     | Project-II                                             |   | 6                      | Project         |
|                        | Total                                                  |   | 15                     |                 |
| SEM 3                  |                                                        |   |                        |                 |
| 11                     | Project-III                                            |   | 9                      | Project         |
| 12                     | Electives (1/2/3/4 credits)                            |   | 6                      | Elective        |
|                        | Total                                                  |   | 15                     |                 |
| SEM 4                  |                                                        |   |                        |                 |
| 13                     | Project-IV                                             |   | 12                     | Project         |
| 14                     | Electives (1/2/3 credits)                              |   | 3                      | Elective        |
|                        | Total                                                  |   | 15                     |                 |
|                        |                                                        |   |                        |                 |
|                        | Total Credits                                          |   | 60                     |                 |



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Semester wise Elective Basket:

## 1. Speech and Natural Language Processing (SNLP)

| SNLP Elective Plan for 15 Credits |                                                                   |                                                     |        |          |
|-----------------------------------|-------------------------------------------------------------------|-----------------------------------------------------|--------|----------|
| Sl no.                            | Course Names                                                      |                                                     | Credit | Type     |
| SEM 1                             |                                                                   |                                                     |        |          |
| 1                                 | Introduction to Speech and Natural Language Processing (Optional) |                                                     | 1      | Elective |
|                                   | Total                                                             |                                                     | 1      |          |
| SEM 2                             |                                                                   |                                                     |        |          |
|                                   | NLP vertical                                                      | Speech Vertical                                     |        |          |
| 2                                 | Syntactic and Semantic Fundamentals of NLP                        | Speech Processing Introduction and Overview         | 2      | Elective |
| 3                                 | Deep Learning                                                     |                                                     | 3      | Elective |
|                                   | Total                                                             |                                                     | 5      |          |
| SEM 3                             |                                                                   |                                                     |        |          |
|                                   | NLP Vertical                                                      | Speech Vertical                                     |        |          |
| 4                                 | Information Retrieval                                             | Speech Recognition and Synthesis: Theory & Practice | 3      | Elective |
| 5                                 | Large Language Model                                              |                                                     | 3      | Elective |
|                                   | Total                                                             |                                                     | 6      |          |
| SEM 4                             |                                                                   |                                                     |        |          |
|                                   | NLP Vertical                                                      | Speech Vertical                                     |        |          |
| 6                                 | Applications of NLP                                               | Speaker and Language Recognition                    | 3      | Elective |
|                                   | Total                                                             |                                                     | 3      |          |
|                                   |                                                                   |                                                     |        |          |
|                                   | Total Credits                                                     |                                                     | 15     |          |



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**Syllabus of all courses of SNLP Basket:**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Credit:</b> 1-0-0-1-1 (L-T-P-S-C)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Course:</b> Introduction to Speech and Natural Language Processing (Optional) |
| <b>Course Type:</b> Elective                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                  |
| <b>Syllabus:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                  |
| <p><b>Unit 1: Lexical Processing in NLP (2 hours)</b><br/>What is NLP &amp; Speech Processing<br/>Text Normalization: Tokenization, Stemming, Lemmatization.<br/>Language Modeling: N-grams, Word2Vec, GloVe.</p> <p><b>Unit 2: Syntactic Processing in NLP (2 hours)</b><br/>Sequence Labeling for Parts-of-Speech (POS) Tagging and Named Entity Recognition (NER).<br/>Context-Free Grammars and Constituency Parsing.<br/>Dependency Parsing.</p> <p><b>Unit 3: Semantic Processing in NLP (3 hours)</b><br/>Word Sense Disambiguation (WSD) – Understanding word meanings in context.<br/>Semantic Role Labeling (SRL) – Assigning roles like agent, object, etc.<br/>Coreference Resolution – Identifying references to the same entity.</p> <p><b>Unit 4: Phonetics &amp; Speech Signal Processing (3 hours)</b><br/>Basics of Speech Production &amp; Phonetics.<br/>Feature Extraction: MFCC, Spectrograms, PLP Features.<br/>Deep Learning for Speech: Introduction to Wave2Vec, WavLM.</p> <p><b>Unit 5: Automatic Speech Recognition &amp; Text-to-Speech (4 hours)</b><br/>ASR Pipeline: Feature extraction → Acoustic modeling → Decoding.<br/>HMM vs. DNN-based ASR systems.<br/>End-to-End ASR Models: Wave2Vec, Whisper API.<br/>TTS Pipeline: Text preprocessing → Prosody → Synthesis.<br/>Deep Learning-based TTS Models: Tacotron, FastSpeech, WaveNet.<br/>Challenges in Speech Synthesis (Low-Resource Languages, Prosody Control).</p> |                                                                                  |
| <b>Text books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                  |
| <ul style="list-style-type: none"><li>Speech and Language Processing – Daniel Jurafsky &amp; James H. Martin (3rd Edition, Draft Available Online)</li><li>Springer Handbook of Speech Processing - Jacob Benesty, M. Mohan Sondhi, Yiteng Arden Huang</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                  |
| <b>Grading Policy:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                  |



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| <ul style="list-style-type: none"><li>• 2 Assignments (14%)</li><li>• 14 Quizzes (42%)</li><li>• 1 End Term (30%)</li></ul> | <ul style="list-style-type: none"><li>• Classroom Notes (5%)</li><li>• Attendance (5%)</li><li>• Activeness in Classes (5%)</li><li>• X-Factor (Originality, Creativity, or Initiative) (5%)</li></ul> |
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| <b>Credit:</b> 2-0-0-0-2 (L-T-P-S-C)                                                                                                                                                                                                                                                   | <b>Course:</b> Syntactic & Semantic Fundamentals of NLP |
| <b>Course Type:</b> Elective                                                                                                                                                                                                                                                           |                                                         |
| <b>Syllabus:</b>                                                                                                                                                                                                                                                                       |                                                         |
| <b>Module 1: Constituency Grammar and Parsing (7 hours)</b><br>Context-Free Grammar (CFG) – definition, rules<br>Parse trees and derivations<br>Ambiguity in parsing (structural, lexical)<br>CYK Parsing Algorithm<br>Probabilistic CFGs (PCFGs)<br>Treebanks and annotated corpora   |                                                         |
| <b>Module 2: Dependency Grammar and Parsing (7 hours)</b><br>Introduction to dependency parsing<br>Dependency relations and tree constraints<br>Transition-based and graph-based parsing approaches<br>Evaluation metrics (UAS, LAS)<br>Neural dependency parsers (e.g., BiLSTM-based) |                                                         |
| <b>Module 3: Semantic Role Labeling and Thematic Roles (7 hours)</b><br>Predicate-argument structure<br>PropBank, FrameNet<br>SRL models: rule-based vs statistical<br>Applications of SRL in QA and IE                                                                                |                                                         |
| <b>Module 4: Word Sense Disambiguation (7 hours)</b><br>Sense inventories<br>Knowledge-based vs supervised vs unsupervised WSD<br>Lesk algorithm, semantic similarity approaches<br>Evaluation frameworks (e.g., SemEval tasks)                                                        |                                                         |
| <b>Text books:</b>                                                                                                                                                                                                                                                                     |                                                         |
| <ul style="list-style-type: none"><li>• Speech and Language Processing – Daniel Jurafsky &amp; James H. Martin (3rd Edition)</li></ul>                                                                                                                                                 |                                                         |
| <b>Grading Policy:</b>                                                                                                                                                                                                                                                                 |                                                         |



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| <ul style="list-style-type: none"><li>• 2 Assignments (14%)</li><li>• 14 Quizzes (42%)</li><li>• 1 End Term (30%)</li></ul> | <ul style="list-style-type: none"><li>• Classroom Notes (5%)</li><li>• Attendance (5%)</li><li>• Activeness in Classes (5%)</li><li>• X-Factor (Originality, Creativity, or Initiative) (5%)</li></ul> |
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| <b>Credit:</b> 2-0-2-0-3 (L-T-P-S-C)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Course:</b> Information Retrieval |
| <b>Course Type:</b> Elective                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                      |
| <b>Syllabus:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                      |
| <p><b>Module 1: Introduction &amp; Classical IR Models (14 hours)</b><br/>Basics of IR, components of IR systems<br/>Boolean and Vector Space Models<br/>TF-IDF and term weighting schemes<br/>Cosine similarity and ranking<br/>Evaluation of IR Systems<br/>Lab: Boolean Retrieval (Lucene or Whoosh) &amp; Building inverted index for TF-IDF scoring &amp; Evaluate a basic retrieval system using TREC-style datasets</p> <p><b>Module 2: Probabilistic IR Models (6 hours)</b><br/>BM25 and Language Models<br/>Query likelihood models<br/>Lab: Implement BM25 scoring and compare with TF-IDF</p> <p><b>Module 3: Web Search and Link Analysis (6 hours)</b><br/>Web crawling and indexing<br/>PageRank and HITS algorithms<br/>Lab: Simulate PageRank for a small web graph</p> <p><b>Module 4: Query Expansion and Relevance Feedback (6 hours)</b><br/>Rocchio algorithm<br/>Pseudo relevance feedback<br/>Lab: Implement Rocchio-based feedback</p> <p><b>Module 5: Neural Information Retrieval (6 hours)</b><br/>Word embeddings in IR<br/>Deep learning models: DSSM, ColBERT, BERT-based retrieval<br/>Lab: Fine-tune a BERT model for document ranking (e.g., MS MARCO)</p> <p><b>Module 6: Trends &amp; Applications (7 hours)</b><br/>Cross-lingual IR, Personalized IR<br/>Legal, medical, scholarly IR systems</p> |                                      |



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| <b>Text books:</b>                                                                                                                                                      |                                                                                                                                       |
| <ul style="list-style-type: none"><li>• Introduction to Information Retrieval, by C. Manning, P. Raghavan, and H. Schütze (Cambridge University Press, 2008).</li></ul> |                                                                                                                                       |
| <b>Grading Policy:</b>                                                                                                                                                  |                                                                                                                                       |
| <ul style="list-style-type: none"><li>• 2 Theoretical Assignments (14%)</li><li>• 2 Practical Assignments (30%)</li><li>• 14 Quizzes (42%)</li></ul>                    | <ul style="list-style-type: none"><li>• Classroom Notes (5%)</li><li>• Attendance (5%)</li><li>• Activeness in Classes (5%)</li></ul> |

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| <b>Credit:</b> 3-0-0-0-3 (L-T-P-S-C)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Course:</b> Applications of NLP |
| <b>Course Type:</b> Elective                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                    |
| <b>Syllabus:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                    |
| <p><b>Module 1: Social NLP (14 hours)</b><br/>Introduction to Social NLP<br/>Preprocessing for Social Text<br/>Sentiment Analysis – Traditional, Neural; Emotion Detection<br/>Applications: Hate Speech and Toxicity Detection, Bias &amp; Fairness in Classification, Fake News and Misinformation, Topic Modeling &amp; Trends, Hashtag and Meme Analysis, Network-aware Text Mining, Cross-lingual and Code-mixed Social Text.</p> <p><b>Module 2: Question Retrieval &amp; Answering (14 hours)</b><br/>Introduction to Question Answering (QA) and Question Retrieval (QR)<br/>Types of QA: factoid, yes/no, causal, multi-hop<br/>Question Retrieval (QR) in QA<br/>QR/QA Datasets: SQuAD, HotpotQA, DROP, MARCO<br/>Evaluation Metrics and Benchmarks<br/>Applications: Expert Finding in Forums, User Reputation and Answer Credibility, Dialogue-based QA.</p> <p><b>Module 3: Educational Technology (14 hours)</b><br/>Introduction to NLP in EdTech<br/>Intelligent Tutoring Systems: Dialogue systems, misconceptions, Socratic questioning<br/>Applications: Question Generation, Answer Evaluation, Text Simplification, Readability Estimation, Feedback Generation, Student Forum Analysis, Learning Analytics, Bias, Fairness, and Explainability.</p> |                                    |
| <b>Text books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                    |
| <ul style="list-style-type: none"><li>• Natural Language Processing with Transformers by Lewis Tunstall, Leandro von Werra, Thomas Wolf (O'Reilly, 2022)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                    |



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- Artificial Intelligence in Education: Promises and Implications for Teaching and Learning by Wayne Holmes et al. (UNESCO Publication)
- "Building Chatbots with Python" by Sumit Raj (Apress)

### Grading Policy:

- |                                                                                                                             |                                                                                                                                        |
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| <ul style="list-style-type: none"><li>• 2 Assignments (14%)</li><li>• 14 Quizzes (42%)</li><li>• 1 End Term (30%)</li></ul> | <ul style="list-style-type: none"><li>• Classroom Notes (10%)</li><li>• Attendance (5%)</li><li>• Activeness in Classes (5%)</li></ul> |
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|------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|
| i.   | Title of the Course                                              | Speech Processing : Introduction & Overview                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |   |   |
| ii.  | Credit Structure                                                 | L                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | T | P | S | C |
|      |                                                                  | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0 | 0 | 4 | 2 |
| iii. | Prerequisite, if any                                             | Exposure to basic concepts in linear algebra and probability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |
| iv.  | Course Content<br><br>(separate sheet may be used, if necessary) | <b>Introduction:</b> Human speech processing, spoken language processing, nature of speech signal, speech signal processing. (3 hrs)<br><br><b>Overview of Tasks:</b> Automatic speech recognition, text to speech synthesis, speaker recognition, language identification and diarization. (3 hrs)<br><br><b>Machine Learning Models for Speech Processing:</b> k-means clustering, Gaussian mixture model (GMM), Hidden Markov Model (HMM) and Neural Networks (NN) for speech processing (9 hrs)<br><br><b>Deep Learning Models for Speech Processing:</b> acoustic modeling using deep feed forward neural network, feature extraction by convolutional neural network, sequence modeling using RNNs and its variants. |   |   |   |   |
| v.   | Texts/References<br>(separate sheet may be used, if necessary)   | <ol style="list-style-type: none"><li>1. L.R. Rabiner and R.W. Schafer, Digital Processing of Speech Signals Pearson Education, Delhi, India, 2004</li><li>2. L. R. Rabiner, B. H. Juang and B. Yegnanarayana, “Fundamentals of speech recognition”, Pearson Education, 2009.</li><li>3. Dong Yu, Li Deng, "Automatic Speech Recognition: A Deep Learning Approach", Springer, 2015</li><li>4. Geoffrey Hinton, et. al., "Deep Neural Networks for Acoustic Modeling in Speech Recognition" IEEE SIGNAL PROCESSING MAGAZINE, Nov 2012.</li><li>5. Articles on Deep Speech Processing</li></ol>                                                                                                                             |   |   |   |   |
| vi.  | Instructor (s)                                                   | S. R. Mahadeva Prasanna                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |   |   |
| vii. | Name of departments to whom the course is relevant               | Elective Course for UG, PG and PhD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |   |   |   |
| viii | Justification                                                    | The course contents include introduction and overview of different topics related to speech processing. The objective is to give conceptual explanations of different tasks and models. Also some coding and hands-on exposure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |   |   |   |





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| ix | Evaluation Scheme | Continuous Evaluation: Earn your credits via investment of your time - deep listening, thinking and practice; Components: asking good questions during or after lectures (10%), MCQs during lectures (10%), weekly assignments (20%), quizzes (best 2 out of 3, 30%), coding and hands-on assignments (30%) |
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|      |                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |   |   |   |
|------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|
| i.   | Title of the Course                                          | <b>Speech Recognition and Synthesis: Theory and Practice</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |
| ii.  | Credit Structure                                             | L                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | T | P | S | C |
|      |                                                              | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0 | 2 | 0 | 3 |
| iii. | Prerequisite, if any                                         | Exposure to basic concepts in linear algebra and probability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |   |   |   |
| iv.  | Course Content<br>(separate sheet may be used, if necessary) | <b>Speech Recognition:</b><br><br><b>Introduction:</b> Speech recognition terminologies and task<br><br><b>Classical Speech Recognition:</b> HMM based speech recognition<br><br><b>Deep Learning based Speech Recognition:</b> DFNN based speech recognition, end-to-end speech recognition, transformer based speech recognition<br><br><b>Speech Synthesis:</b><br><br><b>Introduction:</b> Speech synthesis terminologies and task<br><br><b>Classical Speech Synthesis:</b> Concatenative and HMM based speech synthesis<br><br><b>Deep Learning based Speech Synthesis:</b> Wavenet, tachotron, diffusion based and LLM based models |   |   |   |   |

|       |                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v.    | Texts/References<br>(separate sheet may be used, if necessary) | <p>Speech Recognition:</p> <ol style="list-style-type: none"> <li>1. L. R. Rabiner, B. H. Jhuang and B. Yegnanarayana, "Fundamentals of speech recognition", Pearson Education, 2009.</li> <li>2. Dong Yu, Li Deng, "Automatic Speech Recognition: A Deep Learning Approach", Springer, 2015</li> <li>3. Geoffrey Hinton et.al, "Deep Neural Networks for Acoustic Modeling in Speech Recognition" IEEE SIGNAL PROCESSING MAGAZINE, Nov 2012.</li> </ol> <p>Recent articles on deep speech recognition.</p> <p>Speech Synthesis:</p> <ol style="list-style-type: none"> <li>1. Hunt, A. and Black, A. (1996). "Unit selection in a concatenative speech synthesis system using a large speech database", ICASSP 96, pp 373-376.</li> <li>2. K. Tokuda, et. al., "Speech synthesis based on hidden markov models", Proceedings of IEEE, vol. 101-5, pp. 1234-1252, 2013.</li> <li>3. ISCA Synthesis Interest Group:<br/><a href="https://syntag.org/index.php/Main_Page">https://syntag.org/index.php/Main_Page</a></li> <li>4. CSTR, Simon Kings' page on speech synthesis:<br/><a href="http://www.speech.zone/courses/speech-synthesis/">http://www.speech.zone/courses/speech-synthesis/</a></li> <li>5. CMU Festvox: <a href="http://festvox.org/">http://festvox.org/</a></li> <li>6. HTS web page <a href="http://hts.sp.nitech.ac.jp/">http://hts.sp.nitech.ac.jp/</a> for more information on HMM based speech synthesis system (HTS)</li> <li>7. Google: <a href="https://google.github.io/tacotron/">https://google.github.io/tacotron/</a></li> <li>8. Espnet toolkit link: <a href="https://github.com/espnet/espnet">https://github.com/espnet/espnet</a></li> </ol> |
| vi.   | Instructor (s)                                                 | S. R. Mahadeva Prasanna                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| vii.  | Name of departments to whom the course is relevant             | Elective Course for UG, PG and PhD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| viii. | Justification                                                  | The course contents include introduction and overview of different terminologies and processes related to speech synthesis and speech recognition. The objective is to give hands-on exposure towards development of speech synthesis and recognition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| ix.   | Evaluation Scheme                                              | Continuous Evaluation: Earn your credits via investment of your time - deep listening, thinking and practice; Components: asking good questions during or after lectures (10%), MCQs during                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |



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|  |  |                                                                                                                 |
|--|--|-----------------------------------------------------------------------------------------------------------------|
|  |  | lectures (10%), weekly assignments (20%), quizzes (best 2 out of 3, 30%), coding and hands-on assignments (30%) |
|--|--|-----------------------------------------------------------------------------------------------------------------|

|      |                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |   |   |   |
|------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|
| i.   | Title of the Course                                              | <b>Speaker and Language Recognition: Theory &amp; Practice</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   |   |   |
| ii.  | Credit Structure                                                 | L                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | T | P | S | C |
|      |                                                                  | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0 | 2 | 0 | 3 |
| iii. | Prerequisite, if any                                             | Exposure to basic concepts in linear algebra and probability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |   |   |
| iv.  | Course Content<br><br>(separate sheet may be used, if necessary) | <b>Introduction:</b> Speaker and language recognition terminologies and tasks<br><br><b>Classical Speaker and Language Recognition:</b> GMM and i-vector based speaker and language recognition<br><br><b>Deep Learning based Speaker and Language Recognition:</b> x-vector and other deep architectures based speaker and language recognition and diarization<br><br>Any other important topics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |   |
| v.   | Texts/References<br>(separate sheet may be used, if necessary)   | <ol style="list-style-type: none"><li>1. Homayoon Beigi “Fundamentals of speaker recognition”, Springer 2011.</li><li>2. E. Ambikairajah, et. al., “Language identification: A tutorial,” IEEE Circuits and Systems Magazine, vol. 11, no. 2, pp. 82–108, 2011.</li><li>3. D. A. Reynolds, “Speaker identification and verification using gaussian mixture speaker models,” Speech communication, vol. 17, no. 1-2, pp. 91–108, 1995.</li><li>4. I. Lopez-Moreno, et. al., “On the use of deep feedforward neural networks for automatic language identification,” Computer Speech &amp; Language, vol. 40, pp. 46–59, 2016.</li><li>5. Q. Zhang and J. H. Hansen, “Language/dialect recognition based on unsupervised deep learning,” IEEE/ACM TASLP, vol. 26, no. 5, pp. 873–882, 2018.</li><li>6. B. Jiang, et. al., “Deep bottleneck features for spoken language identification,” PloS one, vol. 9, no. 7, p. e100795, 2014.</li><li>7. D. Snyder, et. al., “Spoken language recognition using x-vectors,” in Odyssey 2018.</li><li>8. Recent articles on deep speaker and language recognition.</li></ol> |   |   |   |   |



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|      |                                                    |                                                                                                                                                                                                                                                                                                             |
|------|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| vi.  | Instructor (s)                                     | S. R. Mahadeva Prasanna                                                                                                                                                                                                                                                                                     |
| vii. | Name of departments to whom the course is relevant | Elective Course for UG, PG and PhD                                                                                                                                                                                                                                                                          |
| viii | Justification                                      | The course contents include introduction and overview of different terminologies and processes related to speaker and language recognition and diarization. The objective is to give hands-on exposure towards development of speaker and language recognition and diarization..                            |
| ix   | Evaluation Scheme                                  | Continuous Evaluation: Earn your credits via investment of your time - deep listening, thinking and practice; Components: asking good questions during or after lectures (10%), MCQs during lectures (10%), weekly assignments (20%), quizzes (best 2 out of 3, 30%), coding and hands-on assignments (30%) |



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## 2. Generative AI

| Generative AI Elective Plan for 15 Credits |                                          |        |          |
|--------------------------------------------|------------------------------------------|--------|----------|
| Sl no.                                     | Course Names                             | Credit | Type     |
| SEM 1                                      |                                          |        |          |
| 1                                          | Introduction to Generative AI (Optional) | 1      | Elective |
|                                            | Total                                    | 1      |          |
| SEM 2                                      |                                          |        |          |
| 2                                          | Reinforcement learning                   | 2      | Elective |
| 4                                          | Deep Learning                            | 3      | Elective |
|                                            | Total                                    | 5      |          |
| SEM 3                                      |                                          |        |          |
| 5                                          | Generative AI Applications               | 1      | Elective |
| 6                                          | Prompt Engineering                       | 2      | Elective |
| 7                                          | Large Language Model                     | 3      | Elective |
|                                            | Total                                    | 6      |          |
| SEM 4                                      |                                          |        |          |
| 8                                          | Agentic AI                               | 3      | Elective |
|                                            | Total                                    | 3      |          |
|                                            |                                          |        |          |
|                                            | Total Credits                            | 15     |          |

### Syllabus:

|                                                    |                                                  |
|----------------------------------------------------|--------------------------------------------------|
| Credit: 1-0-0-0-1 (L-T-P-S-C)                      | Course: Introduction to Generative AI (Optional) |
| Course Type: Elective                              |                                                  |
| Syllabus:                                          |                                                  |
| Unit 1: Overview of AI and Generative AI (5 Hours) |                                                  |



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- Introduction to AI, Machine Learning (ML), and Deep Learning (DL)
- Supervised, Unsupervised and Reinforcement learning
- Neural network as AI model
- Why Learn Generative AI?
- Introduction to Generative AI and its Capabilities
- What Makes Generative AI Different from Traditional AI?
- Evolution of Generative AI

## Unit 2: Tools and Models in Generative AI (5 Hours)

- Overview of Generative Models (GANs, VAEs, Diffusion Models)
- Introduction to Large Language Models (LLMs)
- Tools for Generating Text, Images, Audio, Video, and Code
- Basics of Prompt Engineering

## Unit 3: Applications of Generative AI (4 Hours)

- Real-World Applications of Generative AI Across Domains
- Conversational AI and Chatbots
- Generative AI in Creativity: Image, Music, and Video
- AI for Code Generation and Automation

### Text books:

- Fan, X. (Ed.). (2025). *Learning with generative artificial intelligence: What empirical studies tell us*. Routledge.
- Generative AI for Everyone, Coursera, <https://www.deeplearning.ai/courses/generative-ai-for-everyone/>

### Grading Policy:

- |                                                                                                                    |                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>● 2 Quiz(10%)</li><li>● 1 Mid Sem (20%)</li><li>● 1 End Term (35%)</li></ul> | <ul style="list-style-type: none"><li>● Video Turned on (5%)</li><li>● Attendance (5%)</li><li>● Class end notes (10%)</li><li>● X-Factor (Originality, Creativity, or Initiative) (15%)</li></ul> |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Credit:** 2-0-0-0-2 (L-T-P-S-C)

**Course:** Reinforcement Learning



**Course Type:** Elective

**Syllabus:**

1. Introduction

- Overview and history of reinforcement learning (RL)
- Distinction between RL, supervised, and unsupervised learning
- Components of an RL system: agent, environment, state, action, reward
- Applications of RL in robotics, game playing, resource management
- Markov Decision Processes (MDPs): states, actions, transition probability, rewards, policy

2. Classical Reinforcement Learning Methods

- Dynamic programming and its use in RL
- Monte Carlo methods for policy evaluation
- Temporal Difference (TD) learning: SARSA and Q-learning
- Exploration vs. exploitation: epsilon-greedy, softmax strategies
- Policy improvement and value iteration
- Limitations and challenges with classical approaches

3. Deep Learning Based Reinforcement Learning

- Motivation for deep RL: high-dimensional state spaces
- Deep Q-Networks (DQNs): architecture and training
- Exploration strategies in deep RL
- Challenges: sample efficiency, stability, overfitting

**Text books:**

- Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction* (2nd ed.). MIT Press.
- Li, Y. (2017). *Deep reinforcement learning: Fundamentals, research, and applications*. Springer.

**Grading Policy:**

- |                                                                                                                    |                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>● 2 Quiz(10%)</li><li>● 1 Mid Sem (25%)</li><li>● 1 End Term (35%)</li></ul> | <ul style="list-style-type: none"><li>● Attendance (5%)</li><li>● Class end notes (10%)</li><li>● X-Factor (Originality, Creativity, or Initiative) (15%)</li></ul> |
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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Credit:</b> 3-0-0-0-3 (L-T-P-S-C)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Course:</b> Deep Learning |
| <b>Course Type:</b> Elective                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                              |
| <b>Syllabus:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                              |
| <ul style="list-style-type: none"><li>● <b>Review of Neural networks:</b> Biological and artificial neuron and neural networks, learning &amp; learning laws; feedforward, feedback and hybrid neural networks; back-propagation learning algorithm.</li><li>● <b>Deep feedforward neural networks (DNNs):</b> Training deep network - greedy layerwise training; training belief network, deep feedforward neural network.</li><li>● <b>Convolutional neural networks:</b> CNN vs other neural networks; basic modules of CNN; training and testing of CNN; CNN as feature extractor; CNN as classifier; significance of CNN in object classification task.</li><li>● <b>Sequence models:</b> Sequence models vs discriminative / generate models; Recurrent neural network (RNN); Long short term memory (LSTM) network; gated RNN (GRU); Seq2seq neural network; Attention Modeling; Transformers; applications of sequence models.</li><li>● <b>Generative models:</b> Generative vs discriminative models, variational autoencoder (VAE), generative adversarial networks (GAN), and generative pre-trained transformer (GPT); applications of generative models.</li></ul> <p>Introduction advanced topics in deep learning.</p> |                              |
| <b>Text books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                              |
| <ul style="list-style-type: none"><li>● Michael Nielsen "Neural Networks and Deep Learning" Open Book.</li><li>● Ian Goodfellow, et. al. , "Deep Learning", MIT Press, 2016.</li><li>● Charu Agarwal, "NNDL", Springer 2018.</li><li>● Mithesh M. Khapra, "Deep Learning Part I and II" NPTEL Course</li><li>● Ali Ghodsi, "Deep Learning", University of Waterloo Course</li><li>● C. M. Bishop &amp; H. Bishop, "Deep Learning", Springer, 2024</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                              |
| <b>Grading Policy:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                              |





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Continuous Evaluation: Earn your credits via investment of your time - deep listening, thinking and practice; Components: asking good questions during or after lectures (10%), MCQs during lectures (10%), weekly assignments (20%), quizzes (best 2 out of 3, 30%), End Sem Exam (30%)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Credit:</b> 1-0-0-0-1 (L-T-P-S-C)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Course:</b> Generative AI Applications |
| <b>Course Type:</b> Elective                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                           |
| <b>Syllabus:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                           |
| <ol style="list-style-type: none"><li>Advanced Text Generation and Customization (3 hours)<ul style="list-style-type: none"><li>Fine-tuning and customizing large language models for specific domains</li><li>Prompt engineering and chaining prompts for complex tasks</li><li>Case studies: automated report generation, creative writing, summarization</li></ul></li><li>Image &amp; Multimedia Synthesis (3 hours)<ul style="list-style-type: none"><li>Using GANs and diffusion models for image creation and editing</li><li>Tools for rapid image prototyping and style transfer</li><li>Multimedia content generation: audio, video, and cross-modal use cases</li></ul></li><li>Deployment and Integration (3 hours)<ul style="list-style-type: none"><li>Incorporating generative AI into web and mobile applications (e.g., APIs, plugins)</li><li>Ethics, safety, and bias management in deployment</li><li>Scalability and performance considerations</li></ul></li><li>Hands-On Labs &amp; Project Work (5 hours)<ul style="list-style-type: none"><li>Guided lab sessions using platforms/tools like ChatGPT, DALL-E, Midjourney, Stable Diffusion</li><li>Students design and implement a small project: e.g., an AI-powered content assistant, image generator, or creative application</li></ul></li></ol> |                                           |
| <b>Text books:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                           |



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- [Coursera: Best Generative AI Courses & Certificates](#)
- [Udemy: Top Generative AI \(GenAI\) Courses Online](#)

### Grading Policy:

Continuous Evaluation: Earn your credits via investment of your time - deep listening, thinking and practice; Components: asking good questions during or after lectures (10%), MCQs during lectures (10%), weekly assignments (20%), quizzes (best 2 out of 3, 30%), coding and hands-on assignments (30%)

**Credit:** 2-0-0-0-2 (L-T-P-S-C)

**Course:** Prompt Engineering

**Course Type:** Elective

### Syllabus:

#### 1. Introduction to Prompt Engineering

- Definition, motivation, and importance
- Language model control (temperature, top-p)
- Applications of prompt engineering

#### 2. Elements and Structure of Prompts

- Components: instruction, context, input, output formatting
- Prompt formulation for different tasks

#### 3. Prompting for Tasks

- Summarization, question answering, classification, code generation, reasoning
- Role-play and scenario-based prompts
- Hands-on prompt exercises

#### 4. Advanced Prompt Engineering Techniques

- Few-shot and zero-shot prompting
- Chain-of-thought (CoT) and knowledge-augmented prompting
- Self-consistency, PAL (program-aided LMs), ReAct frameworks



## 5. Safety and Risks

- Prompt injection and security concerns
- Prompt leaking and jailbreaking
- Mitigation strategies

## 6. Future Trends & Resources

- Recent advances and research directions
- Key online resources and guides

### Text books:

- [Prompt Engineering Guide \(dair-ai\)](#)
- Brown et al., “Language Models are Few-Shot Learners”
- [Hugging Face blog: Decoding strategies](#)

### Grading Policy:

Continuous Evaluation: Earn your credits via investment of your time - deep listening, thinking and practice; Components: asking good questions during or after lectures (10%), MCQs during lectures (10%), weekly assignments (20%), quizzes (best 2 out of 3, 30%), coding and hands-on assignments (30%)

**Credit:** 3-0-0-0-3 (L-T-P-S-C)

**Course:** Large Language Models

**Course Type:** Elective

### Syllabus:

#### Unit 1: Foundations of Large Language Models

- **Introduction to Language Models**
  - Fundamentals of language models: History, evolution, and significance in NLP
  - Word Embeddings
  - Neural LMs

- **Transformer Architecture**
  - Transformer basics: Self-attention mechanism, Encoder-Decoder structure
  - Encoder layers and encoder stack
- **Core Components of Transformers**
  - Position encoding, batch normalization, and layer normalization
  - Teacher forcing and masked attention

## Unit 2: Architectures of Large Language Models

- **Decoder-only Large Language Models**
  - Detailed study of GPT architecture (Causal Language Model)
  - Pre-training vs. fine-tuning for downstream applications
  - Decoding strategies: Greedy, Beam Search, Top-k, and Top-p
- **Encoder-only Large Language Models**
  - BERT architecture (Masked Language Model) and training objectives
  - Applications and adaptation for various NLP tasks
- **Tokenization Techniques**
  - Sub-word tokenization, Byte Pair Encoding, WordPiece, and SentencePiece
- **Introduction to Small Language Models**
  - Overview and applications of Small Language Models in resource-constrained settings

## Unit 3: Advanced Architectures and Adaptation of LLMs

- **Encoder-Decoder Models**
  - Introduction to models like BART and T5
  - Understanding the Text-to-Text framework and zero-shot learning
- **Advanced Model Components**
  - Pre-training strategies, scaling laws, and instruction fine-tuning
  - Advanced attention mechanisms and the Mixture of Experts approach
- **Parameter-Efficient Fine-Tuning (PEFT)**
  - Techniques for efficient adaptation and inference
  - Advantages of PEFT in optimizing LLMs for specific tasks
- **Efficient Model Adaptation**
  - Retrieval and tool augmentation for enhanced model capabilities
  - Prompt Engineering, Chains, Memory and Agents

## Unit 4: LLMs in Practice and Future Directions

- **Addressing Challenges in LLMs**
  - Bias, toxicity, hallucination, and alignment in LLMs
  - Interpreting LLMs: Understanding the inner workings and output reasoning
- **Specialized Models and Emerging Trends**
  - Multimodal LLMs
  - Vision-Language models and long-context LLMs
  - Model editing, self-evolving LLMs, and efficient inference
- **Future of LLMs**



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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>○ Ethical implications, evolving capabilities, and emerging applications in LLMs</li></ul>                                                                                                                                                                                                                                          |
| <b>Text books:</b>                                                                                                                                                                                                                                                                                                                                                        |
| <ul style="list-style-type: none"><li>● <a href="#">Introduction to Large Language Models</a></li><li>● <a href="#">GitHub - lcs2-iitd/ELL881-AIL821-2401: An advanced course on LLMs offered at IIT Delhi by Prof. Tanmoy Chakraborty</a></li><li>● Alammar, Jay., Grootendorst, Maarten. Hands-On Large Language Models. United States: O'Reilly Media, 2024.</li></ul> |
| <b>Grading Policy:</b>                                                                                                                                                                                                                                                                                                                                                    |
| Quiz – 10%<br>Project– 30%<br>Midterm examination – 25%<br>End term Examination – 35%                                                                                                                                                                                                                                                                                     |

|                                                                                                                                                                                                                                                 |                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>Credit:</b> 2-0-0-0-2 (L-T-P-S-C)                                                                                                                                                                                                            | <b>Course:</b> <a href="#">Agenti AI</a> |
| <b>Course Type:</b> Elective                                                                                                                                                                                                                    |                                          |
| <b>Syllabus:</b>                                                                                                                                                                                                                                |                                          |
| <b>Unit I: Introduction to Langchan and LangGraph</b>                                                                                                                                                                                           |                                          |
| <b>Unit II: Orchestration using LangChain/LangGraph</b>                                                                                                                                                                                         |                                          |
| <ul style="list-style-type: none"><li>● Components: Chains, Tools, Agents, Memory</li><li>● Integrating external APIs, building retrieval chains</li></ul> <p><b>Practice:</b> Building question-answering systems and knowledge assistants</p> |                                          |
| <b>Unit III: Retrieval-Augmented Generation (RAG)</b>                                                                                                                                                                                           |                                          |
| <ul style="list-style-type: none"><li>● Overview of vector stores (FAISS, Chroma), embedding models</li><li>● Chunking, indexing, and query-time re-ranking</li></ul>                                                                           |                                          |



**Practice:** Building your own Q&A bot using RAG

#### **Unit IV: AI Agents and Multi-Agent Collaboration**

- Task delegation with CrewAI/TinyAgent
- Agent roles, multi-agent planning using MCP (Multi-agent Chain of Prompts)

**Practice:** Implementing a multi-agent system for a real-world task

#### **Unit V: Evaluation and Guardrails**

- Evaluation metrics: fluency, relevance, hallucination
- Implementing safety with Guardrails and structured outputs

**Practice:** Build and test evaluation pipelines using LangChain + Guardrails

#### **Text books:**

Text Books:

1. Documentation from LangChain, OpenAI, CrewAI, Guardrails.ai
2. Harrison Chase, LangChain for LLM Application Development (official tutorials)
3. Alammar, Jay., Grootendorst, Maarten. Hands-On Large Language Models. Japan: O'Reilly Media, 2024.

Reference Books:

. Selected research papers on RAG, LLM Agents, and Evaluation

#### **Grading Policy:**

Quizzes: 5%

Midterm: 15%

End-term: 25%

Project: 35%

Class Participation & Attendance: 20%

### 3. Financial Analytics

| Course Number | Course Title                                   | Credits   | Type     |
|---------------|------------------------------------------------|-----------|----------|
| <b>SEM 1</b>  |                                                |           |          |
| 1             | Introduction to Financial Analytics (Optional) | 1         | Elective |
|               | <b>Total</b>                                   | <b>1</b>  |          |
| <b>SEM 2</b>  |                                                |           |          |
| 2             | Foundation of Financial Analytics              | 2         | Elective |
| 3             | Statistical Methods for Finance                | 2         | Elective |
| 4             | Financial Data Visualization & Dashboards      | 2         | Elective |
|               | <b>Total</b>                                   | <b>6</b>  |          |
| <b>SEM 3</b>  |                                                |           |          |
| 5             | Time Series Analysis & Forecasting             | 2         | Elective |
| 6             | Machine Learning in Finance                    | 2         | Elective |
|               | <b>Total</b>                                   | <b>4</b>  |          |
| <b>SEM 4</b>  |                                                |           |          |
| 7             | Risk Analytics & Portfolio Optimization        | 2         | Elective |
| 8             | FinTech Applications & Ethical AI              | 2         | Elective |
|               | <b>Total</b>                                   | <b>4</b>  |          |
|               |                                                |           |          |
|               | <b>Total</b>                                   | <b>15</b> |          |

Syllabus:

## 1. Introduction to Financial Analytics

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Credit:</b> 1 (1-0-0-0)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Course:</b> Introduction to Financial Analytics (Optional)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Course Description:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| This course provides a comprehensive introduction to the field of Financial Analytics, equipping students with the knowledge and skills to analyze, interpret, and make informed decisions using financial data. It emphasizes the application of data science and artificial intelligence techniques in solving complex financial problems, such as forecasting, portfolio optimization and risk management. Students will learn to work with real-world financial datasets, preprocess and engineer features, apply statistical and machine learning models, and leverage advanced time-series forecasting methods. |
| <b>Learning Objectives:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

- i. **Understand Financial Data:** Identify and analyze various types of financial data, including market, transaction, and alternative data, and comprehend their relevance in financial decision-making.
- ii. **Preprocess and Analyze Data:** Apply data cleaning, transformation, and feature engineering techniques to prepare financial datasets for analysis.
- iii. **Apply Statistical and Machine Learning Models:** Utilize statistical methods and machine learning algorithms to solve financial problems such as risk analysis and portfolio optimization.
- iv. **Perform Time-Series Analysis:** Implement time-series models and forecasting techniques for predicting financial trends, including stock prices, forex rates, and commodity prices.
- v. **Leverage Tools and Technologies:** Use programming languages like Python and R, along with data visualization and big data tools, to analyze and interpret financial data effectively.
- vi. **Solve Real-World Financial Problems:** Develop practical solutions to challenges in algorithmic trading, risk management, and other financial domains through hands-on projects and case studies.

## Syllabus:

### Unit 1: Introduction to Financial Analytics

- Overview of Financial Data Analytics: Importance, scope, and real-world applications in finance.
- Data Sources and Tools: Bloomberg, Yahoo Finance, APIs, Kaggle datasets.
- Fundamentals of Financial Markets: Stocks, bonds, derivatives, and commodities.

### Unit 2: Data Preprocessing and Feature Engineering

- Data Cleaning and Transformation: Handling missing data, outliers, and inconsistencies.
- Feature Engineering: Financial indicators like moving averages, RSI, and Bollinger Bands.
- Time-Series Data: Stationarity, trends, and seasonality.

### Unit 3: Machine Learning in Finance

- Supervised Learning: Regression (Linear, Ridge) and Classification (Decision Trees, Random Forests).
- Unsupervised Learning: Clustering (K-Means, DBSCAN)

### Unit 4: Time-Series Analysis and Applications

- Time-Series Models: ARIMA, ARCH, and VAR for financial data.
- Forecasting Techniques: Exponential smoothing and Holt-Winters method.
- Applications: Stock price prediction and algorithmic trading.

### Unit 5: Ethical and Practical Considerations

- Ethics and Regulations: Financial data privacy, bias in models, and compliance (e.g., GDPR, SEBI).
- Case Studies: Risk analytics and real-world financial decision-making.

## Textbooks:

### • Primary Textbooks

- o *Data Science for Business* by Foster Provost and Tom Fawcett
- o *Forecasting: Principles and Practice* by Robin John Hyndman, George Athanasopoulos
- o *Machine Learning for Asset Managers* by Marcos López de Prado





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- o *Python for Finance: Mastering Data-Driven Finance* by Yves Hilpisch
- **Supplementary Reading**
  - o *Applied Financial Modelling* by William A. Barnett
  - o *The Basics of Financial Econometrics* by Frank J. Fabozzi
  - o *Statistics and Data Analysis for Financial Engineering* by David Ruppert

### Assessment Policy:

- **Assignments (10%):** Regular assignments involving implementation and experimentation with time series models.
- **Quizzes (10%):** Periodic quizzes to test understanding of theoretical concepts.
- **Midterm Exam (30%):** Comprehensive test on foundational concepts and techniques.
- **Endterm Exam (40%):** Comprehensive test on foundational concepts and techniques.
- **Attendance (10%):** Regular attendance is mandatory.

## 2. Foundations of Financial Analytics

|                                                                                                                                                                                                                                                                                                                              |                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Credit:</b> 2 (2-0-0-0)                                                                                                                                                                                                                                                                                                   |                                                                                             |
| <b>Course:</b> Foundations of Financial Analytics                                                                                                                                                                                                                                                                            |                                                                                             |
| <b>Course Description:</b>                                                                                                                                                                                                                                                                                                   |                                                                                             |
| This course introduces the fundamental concepts of financial analytics, covering financial statements, key financial metrics, ratio analysis, and basic financial modeling. Students will learn to analyze financial data using Excel, company reports, and Bloomberg Terminal basics.                                       |                                                                                             |
| <b>Learning Objectives:</b>                                                                                                                                                                                                                                                                                                  |                                                                                             |
| vii.                                                                                                                                                                                                                                                                                                                         | Understand and interpret financial statements (Balance Sheet, Income Statement, Cash Flow). |
| viii.                                                                                                                                                                                                                                                                                                                        | Perform ratio analysis to assess company performance.                                       |
| ix.                                                                                                                                                                                                                                                                                                                          | Work with different financial data types (time series, cross-sectional).                    |
| x.                                                                                                                                                                                                                                                                                                                           | Calculate and interpret key financial metrics (ROI, NPV, IRR, volatility).                  |
| xi.                                                                                                                                                                                                                                                                                                                          | Build basic financial models and forecasts in Excel.                                        |
| xii.                                                                                                                                                                                                                                                                                                                         | Navigate company reports and Bloomberg Terminal for financial data.                         |
| <b>Syllabus:</b>                                                                                                                                                                                                                                                                                                             |                                                                                             |
| <b>Module 1: Introduction to Financial Statements</b>                                                                                                                                                                                                                                                                        |                                                                                             |
| <ul style="list-style-type: none"> <li>Overview of financial analytics</li> <li>Understanding the Balance Sheet (Assets, Liabilities, Equity)</li> <li>Understanding the Income Statement (Revenue, Expenses, Profit)</li> <li>Understanding the Cash Flow Statement (Operating, Investing, Financing Activities)</li> </ul> |                                                                                             |
| <b>Module 2: Ratio Analysis and Financial Health Assessment</b>                                                                                                                                                                                                                                                              |                                                                                             |
| <ul style="list-style-type: none"> <li>Liquidity Ratios (Current Ratio, Quick Ratio)</li> <li>Profitability Ratios (Gross Margin, Net Margin, ROE, ROA)</li> <li>Leverage Ratios (Debt-to-Equity, Interest Coverage)</li> <li>Efficiency Ratios (Inventory Turnover, Receivables Turnover)</li> </ul>                        |                                                                                             |
| <b>Module 3: Financial Data Types and Key Metrics</b>                                                                                                                                                                                                                                                                        |                                                                                             |
| <ul style="list-style-type: none"> <li>Types of financial data: Time Series vs. Cross-Sectional</li> <li>Key financial metrics: ROI (Return on Investment), NPV (Net Present Value), IRR (Internal Rate of Return), Volatility (Risk Measurement)</li> </ul>                                                                 |                                                                                             |
| <b>Module 4: Basics of Financial Modeling and Forecasting</b>                                                                                                                                                                                                                                                                |                                                                                             |
| <ul style="list-style-type: none"> <li>Introduction to financial modeling</li> <li>Building a simple forecasting model (Revenue, Expenses, Profit Projection)</li> <li>Sensitivity analysis using Excel</li> </ul>                                                                                                           |                                                                                             |
| <b>Module 5: Practical Applications and Bloomberg Terminal Basics</b>                                                                                                                                                                                                                                                        |                                                                                             |
| <ul style="list-style-type: none"> <li>Analyzing real-world company reports</li> <li>Introduction to Bloomberg Terminal (Market Data, Financial Statements, Ratios)</li> <li>Case study: Comparing two companies using financial analytics</li> </ul>                                                                        |                                                                                             |
| <b>Tools:</b>                                                                                                                                                                                                                                                                                                                |                                                                                             |
| <ul style="list-style-type: none"> <li>Excel (Formulas, Data Tables, Charts)</li> </ul>                                                                                                                                                                                                                                      |                                                                                             |



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- Bloomberg Terminal
- Python (Yahoo Finance)

### Textbooks:

- Fridson, M. S., and Alvarez, F. (2022). *Financial statement analysis: a practitioner's guide*. John Wiley and Sons.
- Benninga, S., and Mofkadi, T. (2006). *Principles of finance with excel* (Vol. 2). New York, NY: Oxford University Press.

### Assessment Policy:

- Class Participation and Discussions (20%)
- Excel-Based Assignments (20%)
- Case Study Analysis (20%)
- Quiz (10%)
- End Term (30%)

### 3. Statistical Methods for Finance

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| <b>Credit:</b> 2 (2-0-0-0)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Course:</b> Statistical Methods for Finance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Course Description:</b><br>This course introduces essential statistical techniques used in financial analysis, covering descriptive and inferential statistics, probability distributions, hypothesis testing, regression analysis, and Monte Carlo simulations. Students will apply these methods using Excel and Python/R for financial decision-making and risk assessment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Learning Objectives:</b> <ol style="list-style-type: none"> <li>Compute and interpret descriptive statistics (mean, variance, skewness, kurtosis).</li> <li>Analyze relationships between financial variables using correlation and regression.</li> <li>Understand probability distributions (normal, log-normal) in finance.</li> <li>Conduct hypothesis testing for financial data.</li> <li>Apply Monte Carlo simulations for risk assessment.</li> <li>Use Excel and Python/R for statistical analysis in finance.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Syllabus:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Module 1: Descriptive Statistics and Financial Data Analysis</b> <ul style="list-style-type: none"> <li>Measures of central tendency (mean, median, mode)</li> <li>Measures of dispersion (variance, standard deviation, range)</li> <li>Skewness and kurtosis in financial returns</li> </ul> <b>Module 2: Correlation and Dependence in Financial Markets</b> <ul style="list-style-type: none"> <li>Covariance and Pearson/Spearman correlation</li> <li>Autocorrelation in time series data (lag effects)</li> <li>Case study: Correlation between asset classes (stocks, bonds, commodities)</li> </ul> <b>Module 3: Probability Distributions in Finance</b> <ul style="list-style-type: none"> <li>Normal vs. log-normal distributions</li> <li>Fat tails and financial market anomalies</li> <li>Calculating Value-at-Risk (VaR) using distributions</li> </ul> <b>Module 4: Hypothesis Testing and Statistical Inference</b> <ul style="list-style-type: none"> <li>Null vs. alternative hypotheses</li> <li>t-tests, p-values, and confidence intervals</li> <li>Applications in finance (e.g., testing portfolio returns)</li> </ul> <b>Module 5: Regression Analysis for Financial Modeling</b> <ul style="list-style-type: none"> <li>Simple vs. multiple linear regression</li> <li>Interpreting regression coefficients (beta, R-squared)</li> <li>Case study: CAPM (Capital Asset Pricing Model)</li> </ul> <b>Module 6: Monte Carlo Simulations for Risk Assessment</b> <ul style="list-style-type: none"> <li>Introduction to Monte Carlo methods</li> <li>Simulating stock price paths (Geometric Brownian Motion)</li> <li>Estimating portfolio risk with simulations</li> </ul> |
| <b>Tools:</b> <ul style="list-style-type: none"> <li>Excel (Data Analysis Toolpak, Solver)</li> <li>Python (NumPy, pandas, SciPy, statsmodels, matplotlib)</li> <li>R (Tidyverse, quantmod)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Textbooks:</b> <ul style="list-style-type: none"> <li>Hilpisch, Y. J. (2018). <i>Python for finance</i>. O'Reilly Media, Inc.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |



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- Glasserman, P. (2004). *Monte Carlo methods in financial engineering* (Vol. 53). New York: springer.

### Assessment Policy:

- Class Participation and Discussions (20%)
- Excel-Based Assignments (20%)
- Quiz (10%)
- End Term (30%)
- Project (20%)

## 4. Financial Data Visualization and Dashboards

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| <b>Credit:</b> 2 (2-0-0-0)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course:</b> Financial Data Visualization and Dashboards                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Description:</b><br>This course focuses on transforming financial data into impactful visualizations and interactive dashboards. Students will learn best practices in financial data presentation, key performance indicators (KPIs), and dashboard design using Power BI, Tableau, and Python (Matplotlib/Seaborn).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Learning Objectives:</b> <ol style="list-style-type: none"> <li>Apply principles of effective financial visualization (clarity, accuracy, storytelling).</li> <li>Design interactive dashboards for financial KPIs, trends, and benchmarks.</li> <li>Select and visualize financial KPIs (ROI, liquidity ratios, revenue growth).</li> <li>Use Power BI, Tableau, and Python for financial data visualization.</li> <li>Develop dashboards for real-world financial decision-making.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Syllabus:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Module 1: Principles of Financial Data Visualization</b> <ul style="list-style-type: none"> <li>Best practices in financial charts (bar, line, scatter, candlestick)</li> <li>Avoiding misleading visualizations (scaling, distortions)</li> <li>Case study: Effective vs. ineffective financial reports</li> </ul> <b>Module 2: Financial KPIs and Metrics Visualization</b> <ul style="list-style-type: none"> <li>Key financial KPIs (ROI, ROE, EBITDA, liquidity ratios)</li> <li>Time-series trends (revenue, expenses, stock performance)</li> <li>Benchmarking against industry peers</li> </ul> <b>Module 3: Interactive Dashboards in Power BI</b> <ul style="list-style-type: none"> <li>Power BI interface and data import (Excel, SQL, APIs)</li> <li>Designing financial dashboards (filters, slicers, drill-downs)</li> <li>Case study: Portfolio performance dashboard</li> </ul> <b>Module 4: Advanced Dashboarding in Tableau</b> <ul style="list-style-type: none"> <li>Tableau fundamentals (dimensions, measures, calculated fields)</li> <li>Creating dynamic financial dashboards (treemaps, heatmaps)</li> <li>Storytelling with financial data</li> </ul> |
| <b>Tools:</b> <ul style="list-style-type: none"> <li>Python (Matplotlib, Seaborn)</li> <li>Tableau Public</li> <li>Power BI Desktop (DAX formulas, custom visuals)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Textbooks:</b> <ul style="list-style-type: none"> <li>Knaflig, C. N. (2015). <i>Storytelling with data: A data visualization guide for business professionals</i>. John Wiley and Sons.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Assessment Policy:</b> <ul style="list-style-type: none"> <li>Class Participation and Discussions (20%)</li> <li>Lab Exercises (20%)</li> <li>Quiz (10%)</li> <li>End Term (30%)</li> <li>Project (20%)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |



## 5. Time Series Analysis and Forecasting

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| <b>Credit:</b> 2 (2-0-0-0)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Course:</b> Time Series Analysis and Forecasting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Course Description:</b><br>This course provides a comprehensive introduction to time series analysis and forecasting techniques with applications in finance. Students will learn to model trends, seasonality, and volatility in financial data using statistical methods and machine learning approaches. The course emphasizes hands-on implementation using Python (Pandas, Statsmodels) and R.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Learning Objectives:</b> <ol style="list-style-type: none"> <li>Understand fundamental concepts of time series data in finance</li> <li>Apply decomposition methods to identify trends and seasonality</li> <li>Implement ARIMA and GARCH models for forecasting and volatility estimation</li> <li>Utilize moving averages and exponential smoothing techniques</li> <li>Develop forecasts for stock prices, forex rates, and macroeconomic indicators</li> <li>Compare model performance using appropriate metrics</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Syllabus:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Module 1: Introduction to Time Series Analysis</b> <ul style="list-style-type: none"> <li>Characteristics of time series data (stationarity, autocorrelation)</li> <li>Financial time series examples (stock prices, exchange rates)</li> <li>Data visualization techniques for time series</li> </ul> <b>Module 2: Trend and Seasonality Analysis</b> <ul style="list-style-type: none"> <li>Time series decomposition (additive vs. multiplicative)</li> <li>Detrending methods (differencing, transformations)</li> <li>Seasonal adjustment techniques</li> </ul> <b>Module 3: Smoothing Techniques</b> <ul style="list-style-type: none"> <li>Simple and weighted moving averages</li> <li>Exponential smoothing methods (SES, Holt, Holt-Winters)</li> <li>Model selection and parameter tuning</li> </ul> <b>Module 4: ARIMA Models</b> <ul style="list-style-type: none"> <li>Autoregressive (AR) and Moving Average (MA) processes</li> <li>ARIMA model identification and estimation</li> <li>Model diagnostics and validation</li> </ul> <b>Module 5: Volatility Modeling with GARCH</b> <ul style="list-style-type: none"> <li>ARCH and GARCH models for financial volatility</li> <li>Leverage effects and asymmetric volatility</li> <li>Value-at-Risk (VaR) estimation</li> </ul> |
| <b>Tools:</b> <ul style="list-style-type: none"> <li>Python (Pandas, Matplotlib, Statsmodels)</li> <li>R (ggplot2, forecast package)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Textbooks:</b> <ul style="list-style-type: none"> <li>Shumway, R. H., and Stoffer, D. S. (2006). <i>Time series analysis and its applications: with R examples</i>. New York, NY: Springer New York.</li> <li>Hyndman, R. J., and Athanasopoulos, G. (2018). <i>Forecasting: principles and practice</i>. OTexts.</li> <li>Tsay, R. S. (2005). <i>Analysis of financial time series</i>. John Wiley and sons.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |





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## Assessment Policy:

- Class Participation and Discussions (20%)
- Assignments (20%)
- Quiz (10%)
- End Term (30%)
- Project (20%)

## 6. Risk Analytics and Portfolio Optimization

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| <b>Credit:</b> 2 (2-0-0-0)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Course:</b> Risk Analytics and Portfolio Optimization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Course Description:</b><br>This course provides a quantitative foundation for analyzing financial risk and constructing optimal portfolios. Students will learn key risk measures, Modern Portfolio Theory (MPT), and practical optimization techniques using Excel Solver and Python libraries. The course blends theoretical concepts with real-world applications in asset allocation and hedge fund strategies.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Learning Objectives:</b> <ol style="list-style-type: none"> <li>Calculate and interpret key risk metrics (VaR, CVaR, Sharpe ratio)</li> <li>Apply Modern Portfolio Theory to construct efficient frontiers</li> <li>Optimize portfolios using mean-variance analysis</li> <li>Analyze portfolio performance using factor models</li> <li>Implement risk parity and other advanced allocation strategies</li> <li>Use Python and Excel for portfolio optimization tasks</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Syllabus:</b> <p><b>Module 1: Foundations of Risk-Return Analysis</b></p> <ul style="list-style-type: none"> <li>Risk-return tradeoff in financial markets</li> <li>Calculating expected returns and volatility</li> <li>Covariance and correlation matrices</li> </ul> <p><b>Module 2: Risk Measurement Frameworks</b></p> <ul style="list-style-type: none"> <li>Value-at-Risk (VaR) and Conditional VaR (CVaR)</li> <li>Sharpe ratio, Sortino ratio, and other performance metrics</li> <li>Stress testing and scenario analysis</li> </ul> <p><b>Module 3: Modern Portfolio Theory</b></p> <ul style="list-style-type: none"> <li>Markowitz mean-variance optimization</li> <li>Efficient frontier construction</li> <li>Capital Market Line and Tangency Portfolio</li> </ul> <p><b>Module 4: Portfolio Optimization Techniques</b></p> <ul style="list-style-type: none"> <li>Constraints in portfolio optimization</li> <li>Black-Litterman model for incorporating views</li> <li>Monte Carlo simulation for portfolio analysis</li> </ul> <p><b>Module 5: Advanced Topics and Alternative Strategies</b></p> <ul style="list-style-type: none"> <li>Risk parity and minimum variance portfolios</li> <li>Factor investing and smart beta strategies</li> <li>Hedge fund case studies</li> </ul> |
| <b>Tools:</b> <ul style="list-style-type: none"> <li>Python (NumPy, Pandas, PyPortfolioOpt, CVXPY)</li> <li>Excel Solver</li> <li>Bloomberg PORT</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Textbooks:</b> <ul style="list-style-type: none"> <li>Kahn, R., and Grinold, R. (1999). <i>Active portfolio management: A Quantitative Approach for Providing Superior Returns and Controlling Risk</i>. McGraw Hill Professional.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |



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- Hilpisch, Y. J. (2018). *Python for finance*. O'Reilly Media, Inc.

### Assessment Policy:

- Class Participation and Discussions (20%)
- Assignments (20%)
- Quiz (10%)
- End Term (30%)
- Project (20%)



## 7. Machine Learning in Finance

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| <b>Credit:</b> 2 (2-0-0-0)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Course:</b> Machine Learning in Finance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Course Description:</b><br>This course introduces machine learning techniques with applications in finance, covering supervised and unsupervised learning, algorithmic trading strategies, and deep learning for financial forecasting. Students will implement models using Python's Scikit-learn and TensorFlow libraries to solve real-world financial problems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Learning Objectives:</b> <ul style="list-style-type: none"><li>i. Apply supervised learning to credit risk assessment and fraud detection</li><li>ii. Implement unsupervised techniques for customer segmentation</li><li>iii. Develop basic algorithmic trading strategies</li><li>iv. Build LSTM/GRU models for financial time series forecasting</li><li>v. Evaluate model performance using financial metrics</li><li>vi. Address challenges of financial data (imbalanced classes, non-stationarity)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Syllabus:</b><br><b>Module 1: Introduction to ML in Finance</b> <ul style="list-style-type: none"><li>• Overview of ML applications in finance</li><li>• Financial data characteristics and preprocessing</li><li>• Feature engineering for financial problems</li></ul> <b>Module 2: Supervised Learning I - Credit Scoring</b> <ul style="list-style-type: none"><li>• Logistic regression for default prediction</li><li>• Handling class imbalance (SMOTE, class weights)</li><li>• Model interpretation (SHAP, feature importance)</li></ul> <b>Module 3: Supervised Learning II - Fraud Detection</b> <ul style="list-style-type: none"><li>• Anomaly detection techniques</li><li>• Random Forests and XGBoost for fraud classification</li><li>• Evaluation metrics (precision-recall, F1-score)</li></ul> <b>Module 4: Unsupervised Learning - Customer Segmentation</b> <ul style="list-style-type: none"><li>• Clustering algorithms (K-Means, DBSCAN)</li><li>• Dimensionality reduction (PCA, t-SNE)</li><li>• Applications in marketing and risk management</li></ul> <b>Module 5: Algorithmic Trading Basics</b> <ul style="list-style-type: none"><li>• Return prediction models</li><li>• Backtesting framework basics</li><li>• Simple momentum/mean-reversion strategies</li></ul> <b>Module 6: Deep Learning for Financial Forecasting</b> <ul style="list-style-type: none"><li>• Time series forecasting with LSTMs/GRUs</li><li>• Multivariate financial forecasting</li><li>• Sequence modeling challenges</li></ul> |
| <b>Tools:</b> <ul style="list-style-type: none"><li>• Python (TensorFlow, Keras, yfinance, TA-Lib, Backtrader, Scikit-learn)</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Textbooks:</b> <ul style="list-style-type: none"><li>• De Prado, M. L. (2018). <i>Advances in financial machine learning</i>. John Wiley and Sons.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |



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- Jansen, S. (2020). *Machine learning for algorithmic trading*. Birmingham: Packt Publishing.

### Assessment Policy:

- Class Participation and Discussions (20%)
- Assignments (20%)
- Quiz (10%)
- End Term (30%)
- Project (20%)



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### 8. FinTech Applications and Ethical AI

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| <b>Credit:</b> 2 (2-0-0-0)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Course:</b> FinTech Applications and Ethical AI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Course Description:</b><br>This course explores cutting-edge FinTech applications while critically examining the ethical implications of AI in finance. Students will work with blockchain concepts, robo-advisory systems, and regulatory technologies (RegTech), while learning to identify and mitigate bias in financial AI models. The course culminates in a capstone project addressing real-world financial challenges with ESG (Environmental, Social, and Governance) considerations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Learning Objectives:</b> <ol style="list-style-type: none"><li>Understand blockchain fundamentals and applications in finance</li><li>Develop basic robo-advisor algorithms</li><li>Identify and mitigate bias in financial AI models</li><li>Implement ESG metrics in investment strategies</li><li>Build ethical AI solutions using financial APIs</li><li>Navigate regulatory considerations in FinTech</li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Syllabus:</b><br><b>Week 1: FinTech Landscape and Blockchain Basics</b> <ul style="list-style-type: none"><li>Evolution of financial technology</li><li>Blockchain architecture and smart contracts</li><li>Cryptocurrencies vs. CBDCs</li></ul> <b>Week 2: Robo-Advisors and Algorithmic Investing</b> <ul style="list-style-type: none"><li>Portfolio optimization for robo-advisors</li><li>Client risk profiling algorithms</li><li>Backtesting investment strategies</li></ul> <b>Week 3: RegTech and Compliance Automation</b> <ul style="list-style-type: none"><li>AML/KYC technologies</li><li>Transaction monitoring systems</li><li>Regulatory sandboxes</li></ul> <b>Week 4: Ethical AI in Finance</b> <ul style="list-style-type: none"><li>Types of bias in financial models</li><li>Fair lending algorithms</li><li>Model explainability techniques</li></ul> <b>Week 5: ESG and Sustainable Investing</b> <ul style="list-style-type: none"><li>ESG metrics and scoring systems</li><li>Impact investing strategies</li><li>Green fintech applications</li></ul> <b>Week 6: Capstone Project Presentations</b> <ul style="list-style-type: none"><li>Solution demonstrations</li><li>Peer evaluations</li><li>Industry panel feedback</li></ul> |
| <b>Tools:</b> <ul style="list-style-type: none"><li>Python blockchain simulation libraries</li><li>Alpha Vantage API</li><li>Qandle</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



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- Python visualization

### **Textbooks:**

- Chishti, S. and Barberis, J. (2016). *The FINTECH Book: The Financial Technology Handbook for Investors, Entrepreneurs and Visionaries*. John Wiley and Sons.
- O'Neil, C. (2016). *Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy*. United Kingdom: Crown.
- Drescher, D. (2020). *Blockchain basics*. Ascent Audio.

### **Assessment Policy:**

- Ethical Case Analyses (10%)
- Technical Exercises (10%)
- Quiz (10%)
- End Term (30%)
- Capstone Project (40%)